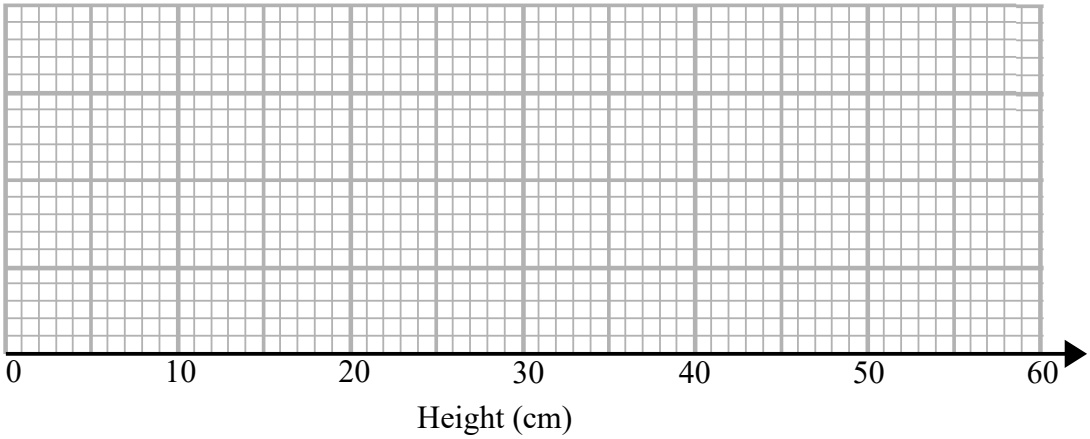
1 The table shows some information about the heights, in cm, of some plants.

Minimum	Lower Quartile	Median	Upper Quartile	Maximum
10	17	24	33	51

Draw a box plot for this information.



(Total for Question 1 is 2 marks)

2 Expand and Simplify $(x - 2)(x + 3)(x - 4)$

…….....

(Total for Question 2 is 3 marks)

3 Write $3.2\dot{4}\dot{5}$ as a fraction in its simplest form.

.....

(Total for Question 3 is 3 marks)

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4 Find the equation of the line perpendicular to $5y - 4x = 6$ which passes through (0,7)

.....

(Total for Question 4 is 2 marks)

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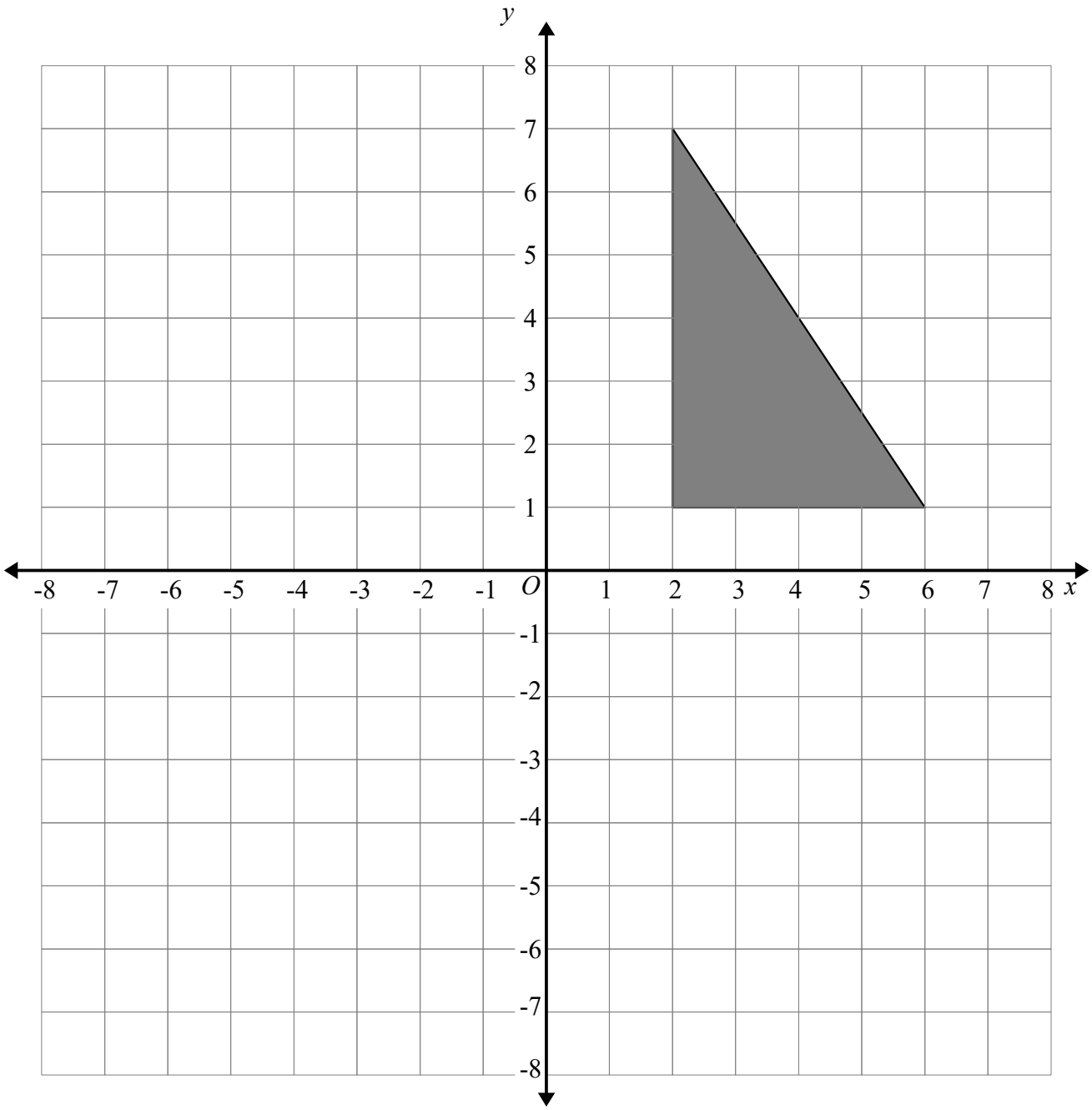
5 Find the value of $(27x^9)^{\frac{2}{3}}$

.....

(Total for Question 5 is 2 marks)

6



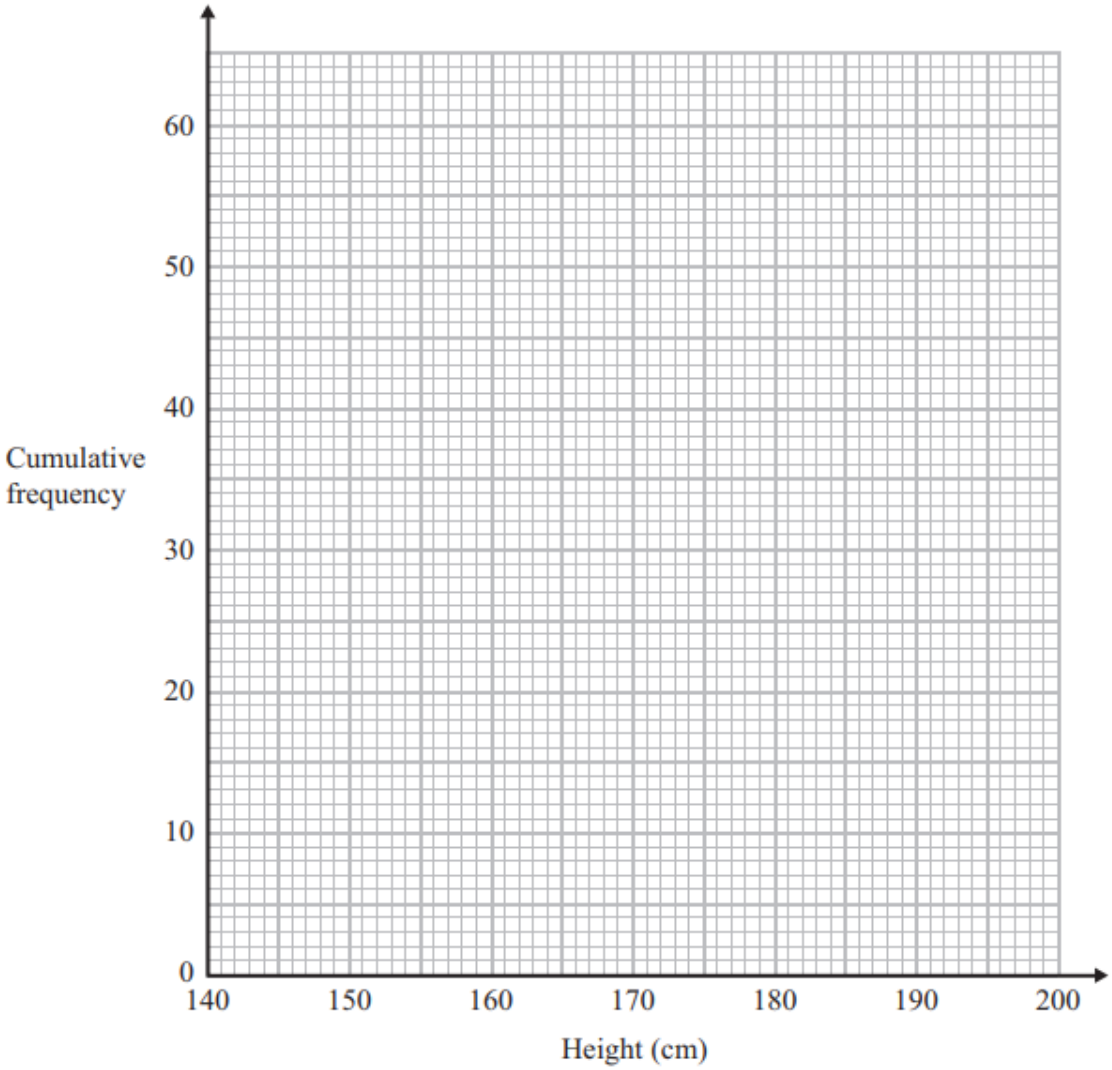
On the grid, enlarge the triangle by scale factor -0.5. centre *O*.

(Total for Question 6 is 2 marks)

7 The cumulative frequency table shows the height, in cm, of some tomato plants.

Height	Cumulative Frequency
$140 < h \leqslant 150$	9
$140 < h \leqslant 160$	19
$140 < h \leqslant 170$	36
$140 < h \leqslant 180$	48
$140 < h \leqslant 190$	55
$140 < h \leqslant 200$	60

(a) On the grid, plot a cumulative frequency graph for this information.



(2)

(b) Estimate the median height.

….....cm
(1)

(Total for Question 7 is 3 marks)

8 Make x the subject of the formula $\frac{a}{b} = \frac{x}{x - 5}$

.....

(Total for Question 8 is 3 marks)

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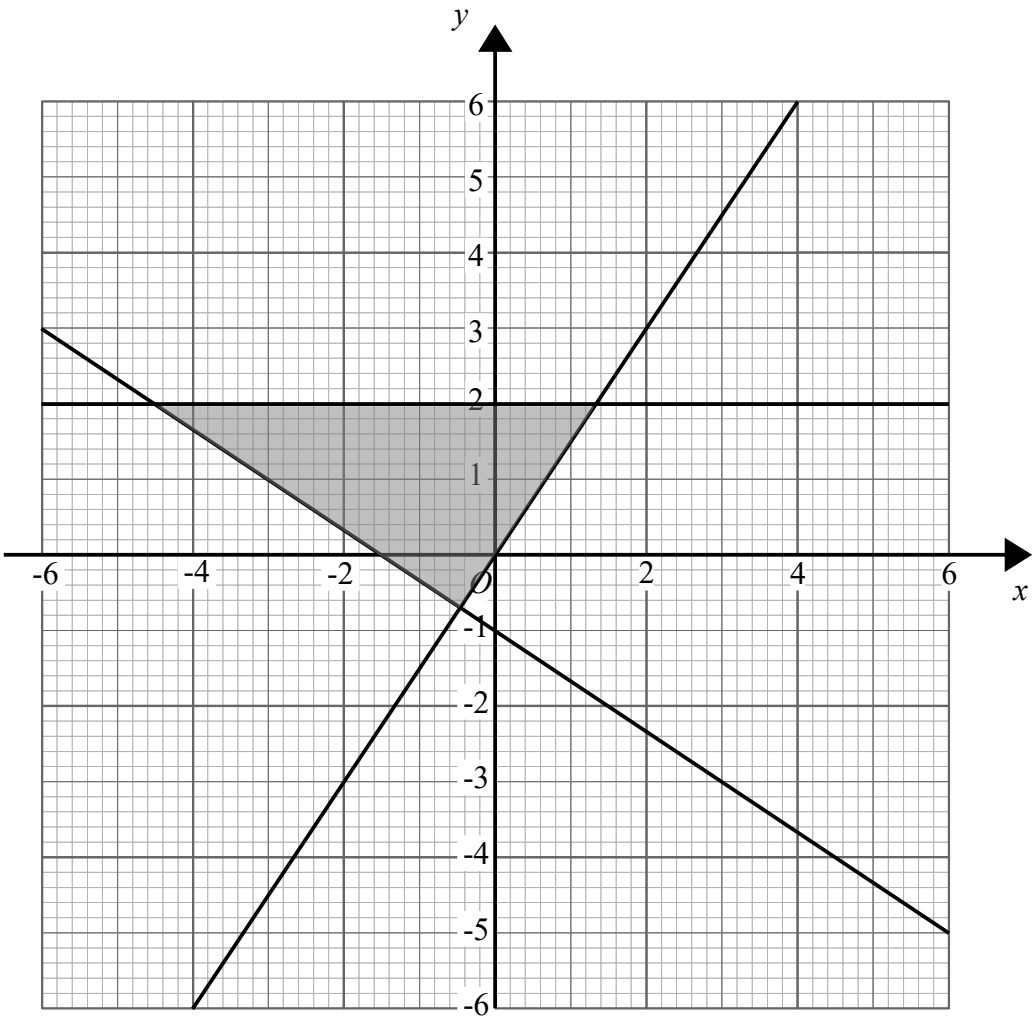
9 Write $(4 - \sqrt{3})^2$ in the form $a + b\sqrt{3}$, where a and b are integers.

.....

(Total for Question 9 is 2 marks)

10



Write down the three inequalities that define the shaded region

.....

.....

.....

(Total for Question 10 is 3 marks)